

Connections AI Solver

Studienarbeit

for the Informatik Department at the
DHBW Heidenheim

by
Ziyi Hong

TODO. TODO 2026

Processing period	January – March 2026, July – September 2026
Student ID	2694933
Course	Allgemeine Informatik
Evaluator	Tobias Heß

Declaration of Independence

I hereby declare that I have written my study on the topic “*Connections AI Solver*” independently and that I have not used any sources or aids other than the indicated ones. I also declare that the submitted electronic version matches the printed version.

MyPlace, August 5, 2026

Place, Date

Signature

Abstract

Contents

List of Tables	VI
1 Introduction	1
1.1 Structure	1
2 Foundations	2
3 Related Work	9
4 Experimental Design and Implementation	15
5 Evaluation and Discussion	16
5.1 Experiments	16
6 Conclusion and Future Work	17
Bibliography	IV

Contents

CoT Chain-of-Thought

CSP Constraint Satisfaction Problem

List of Tables

2.1	Relation types in NYT Connections, with illustrative examples.	3
-----	--	---

1 Introduction

NYT Connections asks players to sort sixteen words into four groups of four. Each group carries one of four difficulty tiers, ranging from yellow (easiest) to purple (hardest). Published daily, the puzzle has become a popular benchmark for testing the reasoning abilities of language models. Two challenges make Connections difficult to solve.

The first challenge is lexical or semantical. Solving Connections requires several distinct kinds of language understanding at once. Some groups rest on straightforward synonymy. Others depend on shared themes, wordplay, or knowledge of pop culture and named entities. A single puzzle typically mixes several of these relation types, and is deliberately constructed so that some words could plausibly belong to more than one candidate group under a superficial reading, an effect commonly referred to as a red herring.

The second challenge is structural. Because a word cannot be reused across groups, correctly identifying even one group depends not only on recognizing a plausible shared category among four words in isolation, but also on the arrangement of the remaining words. A locally plausible grouping can still be wrong if it leaves the remaining words unable to form additional valid groups of four.

In previous studies, the first challenge has been widely discussed, through embedding-based clustering or large language models to propose or judge candidate groupings Section 3.1. This thesis evaluates how far such approaches can be pushed, including several forms of knowledge injection beyond raw embeddings, and finds that they consistently fall short in similar ways (note: to be written, do the reference later!!!). The second challenge, ensuring that all sixteen words are partitioned simultaneously and exclusively, has received comparatively little attention on its own. Taking inspiration from solving Wordle algorithmically with information theory [1], this thesis treats the puzzle’s interactive structure, submitting a guess, observing feedback, and revising, as central to addressing this second challenge, and asks how much of the puzzle’s difficulty it can resolve by algorithm.

1.1 Structure

2 Foundations

This chapter establishes the conceptual and technical foundation for this thesis. Section 2.1 introduces the NYT Connections puzzle and its rules. Section 2.2 develops a taxonomy of the semantic relations underlying puzzle categories and links it to the puzzle’s difficulty structure. Section 2.3 discusses why the puzzle is difficult for automated solvers in general. Section 2.4 motivates the focus on small, locally deployable language models. Sections 2.5 and 2.6 cover the two technical building blocks of the proposed solver – text embeddings/clustering and LLM prompting strategies. Section 2.7 reviews prior solver designs for Connections and positions this thesis relative to them.

2.1 The NYT Connections Puzzle

NYT Connections is a daily word puzzle in which players must partition a set of sixteen words into four disjoint groups of four. Each correct group is assigned one of four color-coded difficulty labels – yellow (easiest), green, blue, and purple (hardest) – reflecting the puzzle’s intended difficulty in recognizing the underlying grouping relation.

Two evaluation protocols are relevant for this thesis and are used interchangeably across the experiments described in ??¹:

- An **interactive setting** with a limited number of attempts (four in the original game) and “one-away” feedback: after each submitted group, the player learns whether the group was fully correct, wrong, or contained exactly three correct words (“one away”).
- An **all-at-once setting** that requires predicting the complete 4×4 partition in a single submission, without intermediate feedback.

Solver communities have also adopted self-imposed variants to increase difficulty, such as attempting the purple category first (“purple-first”), which reverses the typical order of play by prioritizing the most abstract relation before the easier categories are identified. This practice illustrates that even human solvers treat the difficulty ordering yel-

¹Adjust the chapter label `ch:methodology` once your methodology chapter exists.

low $\rightarrow$ green $\rightarrow$ blue $\rightarrow$ purple as a meaningful, exploitable structure rather than an arbitrary label – a point taken up again in Section 2.2.

2.2 Relation Types and the Structure of Difficulty

The categories in a Connections puzzle are not homogeneous in the kind of linguistic or world knowledge they require. Following the taxonomy summarized in Table 2.1, four broad relation types can be distinguished, with a fifth “mixed” category covering cases that combine more than one type.

Table 2.1: Relation types in NYT Connections, with illustrative examples.

Relation type	Example category and words
Semantic similarity	<i>FURTHEST POINT</i> : END, EXTREME, OPPOSITE, POLE [2]
Semantic relatedness	<i>THINGS YOU CAN DO ON SOCIAL MEDIA</i> : COMMENT, LIKE, LURK, POST [2]
Form- or rule-based relations	<i>ART MOVEMENTS, WITH “-ISM”</i> : BRUTAL, IMPRESSION, MANNER, REAL [2]
Referential relations	<i>RIHANNA #1 HITS</i> : DIAMONDS, SOS, UMBRELLA, WORK [3]
Mixed categories	<i>CAR BRAND HOMOPHONES</i> : INFINITY, MINNIE, OPAL, OUTIE [4]

Semantic similarity relations rest on near-synonymy or shared hypernymy (e.g., “types of flowers”); semantic relatedness relations hold between words that are not strictly similar but co-occur within a common theme or script; form- or rule-based relations are determined by surface regularities such as spelling, morphology, or conventional symbol meanings; referential relations require external world knowledge about named entities or factual lists. The hardest (purple) categories draw disproportionately on the less standard relation types – word and letter patterns such as rhymes, homophones, anagrams, or palindromes, unmarked fill-in-the-blank phrases, and pop-culture or proper-noun knowledge – and frequently combine more than one relation type at once [5].

Working hypothesis adopted in this thesis. The color-coded difficulty ordering correlates, at least approximately, with this relation-type taxonomy: yellow and green categories tend toward semantic similarity and relatedness, which are directly addressable by distributional/embedding-based methods, whereas blue and purple categories skew toward form-based, referential, and mixed relations, which embeddings alone are ill-suited to capture and which instead require either symbolic pattern matching or LLM-internal world knowledge. This mapping is not drawn from a specific external source but is proposed here as an explanatory lens; it is revisited in ??² as the basis for the per-tier error analysis.

2.3 Challenges for Automated Solvers

The relation-type diversity described above already indicates why Connections is difficult for automated approaches: a single puzzle may combine semantic similarity, semantic relatedness, form- or rule-based cues, and referential knowledge. Consequently, the task spans multiple NLP subproblems simultaneously – distributional semantics and sense disambiguation for similarity/relatedness categories, pattern matching for form-based categories, and retrieval of external, often culturally specific knowledge for referential categories – so that no single notion of similarity is sufficient across all groups.

The puzzle is also difficult by design. Categories frequently overlap in plausibility, creating red herrings: words that appear to fit one candidate grouping but ultimately belong elsewhere. Context clues are consequently critical for disambiguating individual words, and because the four groups must jointly partition all sixteen words, early (incorrect) commitments constrain what remains achievable for the rest of the puzzle [5].

Loredo Lopez, McDonald, and Emami [6] explicitly design their Connections benchmark to penalize fast, superficial-cue-driven (“System-1”) guesses. This framing aligns with the shortcut-learning perspective: models can succeed by exploiting easy-to-use regularities rather than recovering the intended global grouping structure, a behavior expected whenever the data distribution offers shortcut opportunities and the evaluation objective rewards whatever features are most locally predictive [7]. Connections’ daily release cadence additionally supports robust, leakage-resistant model comparison, since each day’s

²Adjust once your results chapter exists.

puzzle is unseen at evaluation time relative to a model’s training cutoff [6].

2.4 Small and Resource-Constrained Language Models as Solving Agents

Most reported LLM results on Connections-style benchmarks use large, cloud-hosted models [8, 6, 9]. This thesis instead centers on small, locally executable models in the 7–8B parameter range (e.g., Mistral 7B, Llama 3.1 8B), run through a local inference server (Ollama) rather than a hosted API. Three considerations motivate this focus:

1. **Deployment constraints.** Local execution removes dependence on external API availability, per-token cost, and data egress, which is relevant whenever a solver is meant to run on constrained or offline infrastructure.
2. **The capability gap as the actual object of study.** Because small models trail larger ones on complex reasoning tasks, they expose failure modes – shortcut reliance, weak constraint tracking across a 16-word global partition, limited world knowledge for referential categories – more readily than large models do. This makes small models a more informative testbed for evaluating whether *external scaffolding* (embeddings, clustering, structured verification) can substitute for missing model-internal capability, which is the central question this thesis investigates.
3. **Comparability to prior work.** Loredó Lopez, McDonald, and Emami [6] already evaluate multiple LLMs of varying size under different prompting schemes on Connections, which provides a point of comparison for the relative performance of the small models used here (Section 2.7).

2.5 Text Embeddings and Clustering for Lexical Grouping

At the representation level, Connections is closely tied to text embeddings, semantic similarity, and clustering. Early work on distributed word representations introduced dense word vectors capturing lexical regularities through co-occurrence statistics [10]. Subsequent surveys trace the shift from such static word vectors to contextualized representations, including sentence-level embeddings optimized for semantic similarity [11];

transformer-based embedding models are now the standard approach for encoding context-dependent lexical and semantic relations [12], and a broader survey perspective summarizes representation choices and their implications for similarity modeling across NLP tasks more generally [13].

Classical clustering algorithms – k-means, k-means++, and DBSCAN, together with keyword-clustering techniques – group items based on distances within such embedding spaces [14, 15]. More recent work shows that LLM-derived embeddings can improve resulting cluster structure, and that clustering itself can be reframed as an in-context classification problem solved directly by an LLM rather than by a separate distance-based algorithm [16, 17].

This thesis uses SentenceTransformer-style encoders (specifically `BAAI/bge-large-en-v1.5`, alongside comparison runs with other encoder families such as the E5 series and MiniLM) as the embedding backbone for the non-LLM baseline and for the clustering pre-processing stage of the hybrid pipeline (??). The choice of embedding family is treated as an experimental variable rather than fixed a priori, following the observation in Section 2.7 that prior work on Connections instantiates broadly comparable solver structures with substantially different embedding families, without directly comparing them under a controlled setup.

2.6 Prompting Strategies for LLM Reasoning

Chain-of-thought (CoT) prompting elicits intermediate reasoning steps from an LLM before it commits to a final answer and has been shown to improve performance on complex, multi-step tasks [18]. Prompting-technique reference guides collect CoT and related variants in a practitioner-oriented form [19].

The solver pipelines evaluated in this thesis instantiate several points along this spectrum: a zero-shot condition, a few-shot condition using a single worked example, and a CoT condition that decomposes reasoning into an explicit analyze–group–verify sequence (see `system_prompt()` in the implementation). A self-consistency variant, which samples multiple reasoning traces and aggregates over them rather than committing to a single greedy completion, is evaluated as an additional condition. Loredo Lopez, McDonald, and Emami [6] similarly contrast direct input–output prompting against CoT and

self-consistency variants on Connections specifically, which this thesis uses as a point of reference in Section 2.7 and ??.

2.7 Related Work on NYT Connections Solvers

Three prior studies directly inform the solver design used in this thesis.

Todd et al. [8] study Connections using both embedding-based heuristics and prompted LLMs. Their non-LLM baseline encodes words with SentenceTransformer-style models, exhaustively scores all $\binom{16}{4}$ candidate four-word groups by within-group cosine-similarity cohesion, and constructs a solution iteratively; they also consider an all-at-once variant that ranks complete 4×4 partitions by summed cohesion.

Loredo Lopez, McDonald, and Emami [6] propose a “System-1-like” heuristic baseline using Multilingual-E5-Large-Instruct embeddings, scoring candidate groups with a discriminative function

$$S = G - P, \tag{2.1}$$

where G aggregates multiple within-group cohesion signals and P penalizes groups that are insufficiently distinctive from the remaining words. Full solutions are assembled via beam search. They evaluate multiple LLMs under direct, chain-of-thought, and self-consistency prompting schemes.

Correspondence to this project’s implementation. The `score_group()` function in `herustic_solver.py` computes

$$G = 0.4 \cdot I + 0.3 \cdot s + 0.3 \cdot V$$

(a weighted combination of negative inertia, minimum pairwise cosine similarity, and a mean/variance cohesion term) and a final score $S = G - P$, assembled via a beam-search procedure over group choices – structurally identical to the scoring function in Equation 2.1 as described by Loredo Lopez et al. [6].

This correspondence must be stated explicitly in the methodology chapter as the basis for the reproduced baseline, both to satisfy the “solides Fundament” principle of the DHBW guideline and to avoid any appearance of an unattributed structural borrowing.

Zhang, Jiang, and Ye [9] frame the task as equal-size constrained clustering and apply Balanced k-means on static word embeddings (GloVe/word2vec), with an alternative KMeans-plus-rebalancing baseline to enforce the four-groups-of-four partition. They additionally explore LLM-based pipelines including in-context learning, parameter-efficient fine-tuning (QLoRA), rationale distillation from a larger teacher model, and feedback-driven refinement.

Synthesis and positioning of this thesis. These three studies converge on a common structural insight – that a Connections solver decomposes into (a) a representation/scoring stage over candidate groups and (b) an assignment stage that resolves the equal-size global partition – while differing substantially in how each stage is instantiated: embedding family (E5 [6] vs. SentenceTransformer/MPNet [8] vs. static word2vec/GloVe [9]), scoring function, and assignment method (iterative greedy construction [8], beam search [6], balanced/rebalanced k-means [9]). None of the three directly compares embedding families under an otherwise fixed solver structure, and only Zhang, Jiang, and Ye [9] treat the assignment stage as an explicitly constrained combinatorial problem – but resolve it with a clustering heuristic rather than an exact method.

This thesis addresses both gaps directly. First, it reproduces the [6] scoring function as a fixed non-LLM baseline and instantiates it with multiple embedding families under an otherwise controlled setup (??), extending the embedding-family comparison that [8, 6, 9] motivate but do not themselves carry out. Second, building on the constrained-clustering framing of [9], it replaces the balanced/rebalanced clustering heuristic with exact and near-exact constrained-assignment methods (dynamic programming and integer linear programming via OR-Tools) applied to LLM- and embedding-scored candidate groups, combined with an explicit LLM-based trap/red-herring verification stage – directly addressing the red-herring challenge identified in Section 2.3 [5]. Third, unlike [8, 6, 9], which evaluate primarily at the level of overall puzzle success, this thesis evaluates performance separately per difficulty tier (Section 2.2), which is necessary to assess the specific research question of whether the proposed hybrid pipeline improves solving accuracy on the easier (yellow/green) categories relative to the small-model baselines established in Section 2.4.

3 Related Work

This chapter provides an overview of prior work relevant to this thesis, including work that directly solves Connections using LLMs and work that inspired the methods used in this study. It first introduces prior work on solving Connections with AI. It then presents related work on the methods used, which serves as the knowledge base for this study.

3.1 Prior Work on Connections Solving

Current work on this topic mainly uses LLMs as the main solution for solving Connections. These studies compare how well different models solve Connections. Some use Connections as a benchmark to test LLM capability, while others explore different prompting techniques and methods. Todd et al. use three methods, namely a sentence-embedding baseline, a comparison between two LLMs (GPT-3.5-Turbo-1106 and GPT-4-1106-Preview), and Chain-of-Thought (CoT) prompting. Their experiments allow five incorrect and five invalid guesses per puzzle, which departs from the original game’s rules. GPT-4 Turbo with CoT achieves the highest success rate at 58.11%. The best-performing sentence-embedding baseline (MPNet) reaches 11.6%. LLMs are often stumped by categories that involve non-semantic word properties, abstract features, or context-dependent usage. They also report a finding relevant to this thesis. Puzzle order matters, since presenting puzzles in their original, grouped order achieves substantially higher accuracy than presenting them in randomized order [20].

A similar work from Samadarshi et al. benchmarked several LLMs against both novice and expert human players and creates a taxonomy of knowledge types required to solve connections. They used in total 438 Connections puzzles and their best performing models Claude Sonnet 3.5 achieved 18% success rate of fully solving the puzzles. They categorizes NYT Connections categories along two main dimensions, word form (covering morphology, orthography, phonology, and multiword expressions) and word meaning (covering semantic relations such as synonymy, polysemy, and hypernymy, along with associative and encyclopedic relations), with a further category for groups that combine both form-based and meaning-based cues. Results show that models performed comparably well on semantic relation categories but significantly worse on the other types.

3 Related Work

More recent work from Loredó Lopez, McDonald, and Emami compare three game configurations: single attempt, four tries without feedback (one away prompt) and four tries with feedback. Human solvers outperform LLM under all three configurations. The best performing model Claude 3.5 under full hints reaches 40.25%, while the human full-hints score reaches 60.67%. They also experiment on different prompting techniques. CoT and self-consistency prompting yield diminishing or even negative returns as puzzle difficulty increases. Simple input-output prompting sometimes outperforms these more elaborate reasoning strategies. Their heuristic (non-LLM) baseline, using Multilingual-E5-Large-Instruct embedding and beam search, reaches 13.25% accuracy.

Zhang, Jiang, and Ye compare in-context prompting of a large model (DBRX) against fine-tuning and distilling a smaller model (Mistral 7B) on solved puzzles. They found that fine-tuning a smaller model (Mistral 7B) directly on solved puzzles fails to capture the task itself, which underperformed even their clustering baseline. Their strongest result comes from Verbal Reinforcement Learning, in which a larger supervisor model critiques the fine-tuned model’s output in natural language across three rounds, roughly doubling the average per-puzzle match rate relative to single-pass prompting [23].

3.2 Word Puzzles as Weighted Constraint Satisfaction Problem (CSP)

CSPs are mathematical questions defined as a set of objects whose state must satisfy a number of constraints or limitations [24]. A (finite domain) CSP can be expressed in the following form: given a set of variables, together with a finite set of possible values that can be assigned to each variable, and a list of constraints, find values of the variables that satisfy every constraint [25]. It is formally defined as a triple $\langle X, D, C \rangle$ where X is a set of variables, D specifies a domain of possible values for each variable, and C is a set of constraints restricting which combinations of value assignments are permitted [24].

In artificial intelligence, a Weighted Constraint Satisfaction Problem, also known as Valued Constraint Satisfaction Problem, is a constraint satisfaction problem in which preferences among solutions can be expressed [26]. A plain CSP, treats every solution as equally good: it either satisfies the constraints or it does not. Connections needs more than that. Not every four-groups-of-four split is a correct solution, only one is. What is available

instead is a measure of how well four words seem to belong together, based on their similarity. Thus, Connections can be defined as a weighted CSP. It keeps the hard constraint (a valid four-groups-of-four partition) but adds a score to each candidate value assignment, so that solving the problem means finding the constraint-satisfying assignment with the best total score, not just any assignment that satisfies the constraints.

3.2.1 Solving Word Puzzles as Weighted CSP

Applying this concept to word puzzles is not new. Ginsberg formalizes American-style crossword solving as a weighted CSP in the program DrFill: each grid cell is a variable, each candidate fill is a value, and the constraints require intersecting words to agree on shared letters [27]. Because many fills are plausible for a given clue, each candidate fill is given a likelihood-based weight, and Dr.Fill searches for the assignment that maximizes total weight while still satisfying the grid's hard constraints. It draws candidate fills from several sources: a database of over 47,000 previously published crosswords with 1.9 million unique clues, two dictionaries of varying size, 1.2 million synonyms drawn from WordNet and other online sources, and a database of Wikipedia titles and word pairs. Each candidate fill is then scored on five criteria: whether the clue matches one seen before in the clue database, whether the fill's part of speech agrees with the part of speech inferred from the clue, the fill's crossword "merit" (a hand-rated measure of how desirable a word is as a crossword answer in general), whether the fill is an abbreviation, and whether the clue is a fill-in-the-blank type resolvable against Wikipedia text. Dr.Fill then searches for the assignment that maximizes total score while still satisfying the grid's hard letter-intersection constraints. Two further design choices in Dr.Fill are worth noting. First, they found that looking further ahead before committing to a choice gave no real benefit, and the system used only a single step of lookahead instead. Second, branch-and-bound, a standard pruning technique for weighted CSPs, was tested and dropped because it added little value in this domain, as a result postprocessing was used instead. In both cases, a more sophisticated search method did not beat a simpler one, once the scoring function and hard constraints were doing most of the work. Since complete knowledge of what each clue means is impractical to obtain, Dr.Fill relies mainly on the hard constraint that crossing words must share letters to narrow down candidates, not on understanding the clues themselves. Dr.Fill inspired the methodology used in this thesis, which also

treats Connections as a weighted CSP and uses a scoring function to evaluate candidate groupings.

3.2.2 Set Partitioning

The constraint C of Connections (every word belongs to exactly one group and every group contains exactly four words), more precisely speaking, is a combinatorial optimization problem known as set partitioning. A partition of a set is a way of dividing all of its elements into non-empty, pairwise disjoint subsets whose union is the original set [28]. The set partitioning problem asks, given a ground set and a collection of candidate subsets each associated with a cost, to select a subset of these candidates that forms a partition of the ground set at minimum or maximum total cost [29]. The sixteen Connections puzzle words are the ground set, candidate four-word groups are subsets among which a selection must be made, and each candidate group's score from the weighted CSP explained above serving is its cost. Identifying this structure is what motivates the grouping strategy adopted later in this thesis.

3.3 Information Theory

Information theory is the mathematical study of the quantification, storage, and communication of a particular type of mathematically defined information [30]. It gives information a precise mathematical definition grounded in probability [31]. Gaining information and reducing uncertainty about a random event are the same thing. By convention, information is measured in base-2 logarithms, giving the bit as its unit: each time half of the probability mass is ruled out, exactly one bit is gained [32]. The information contained in an outcome with probability $p(x)$ is $\log_2 \frac{1}{p(x)}$.

The probability-weighted average of this information across all outcomes is the entropy, denoted $H(X)$:

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{1}{p(x)}. \quad (3.1)$$

Entropy serves as the connection between probability and information content. It measures how uncertain a random event is on average. Higher entropy means more uncertainty to begin with, so more information must be acquired to become certain about the outcome. Entropy is maximized when probability is spread evenly over all outcomes. In that

3 Related Work

case it reduces to

$$H_{\max}(\mathcal{X}) = \log_2 |\mathcal{X}|, \quad (3.2)$$

where $|\mathcal{X}|$ is the number of possible outcomes [32].

Conditional entropy, $H(X | y)$, measures how much uncertainty about X remains after observing an outcome y . The reduction in entropy achieved by an observation is the information gain. Taken in expectation over all possible outcomes of an observation, before that observation is actually made, this gives the expected information gain of an action:

$$IG(\text{guess}) = H(X) - \mathbb{E}[H(X | \text{feedback})]. \quad (3.3)$$

This quantity is what solvers use to decide which action is most useful before knowing what the feedback will be. An action with higher expected information gain is expected to narrow down the remaining candidates more, regardless of which specific feedback is eventually returned.

3.3.1 Solving Wordle with Information Theory

Wordle is another word puzzle game from NY Times. It asks the player to guess a five-letter word within six attempts. After each guess, every letter is marked as correct and in the right position, present but in the wrong position, or absent, giving detailed per-letter feedback. Information theory based Wordle solvers apply this framework directly: at each step, the candidate word maximizing expected information gain is chosen as the next guess [33]. According to Grant Sanderson pointed out in his video, the best possible first guess yields an expected information gain of 5.89 bits, reducing the remaining uncertainty by roughly half in a single observation [1]. Aladaileh et al. report that entropy-based selection outperforms a heuristic that selects words by letter distribution, and propose the approach as a general baseline for applying Shannon entropy to other games [33]. Information theory can be applied to Connections as well. Detailed description see next chapter.

3.4 Strategies for Puzzle Games

The previous sections introduced the structural theory, and then the information theory as a way of measuring how much a single observation can reveal. After measuring information, a solver needs a rule that turns the current state of knowledge into the next

3 Related Work

move. This section covers the other question: given feedback from previous attempts, how should the next guess be chosen?

Cunanan and Thielscher [34] address this question by defining a general class of games to which Wordle belongs. A guessing game is a tuple (G, S, R, r^*, a) , where G is the set of allowable guesses, S the set of possible secrets, R the set of possible responses, r^* the response indicating the secret has been found, and $a : G \times S \rightarrow R$ the answering function that determines which response a guess receives. After a sequence of guesses and responses, the candidate set is the set of secrets consistent with every response observed so far, and a strategy is a rule mapping each candidate set to the next guess to submit.

Within this framework, strategies are expressed as valuations: functions that score each possible guess against the current candidate set. Cunanan and Thielscher collect several valuations previously developed for Mastermind, including minimising the size of the largest remaining candidate set, maximising information gain in the sense of the previous section, and maximising the number of distinct responses a guess can produce. Evaluating these on Wordle, they find that none performs best alone, and that maximising information gain in particular does not yield the optimal strategy [34].

4 Experimental Design and Implementation

Connections maps onto this triple directly. Let $W = \{w_1, \dots, w_{16}\}$ be the sixteen puzzle words. Each word gets its own variable, so $X = \{x_1, \dots, x_{16}\}$, and x_i is the group assigned to word w_i . Every variable has the same domain, $D_i = \{1, 2, 3, 4\}$, one value for each of the four groups. The constraint set C requires each group to contain exactly four words:

$$C : \quad |\{i : x_i = g\}| = 4 \quad \text{for every } g \in \{1, 2, 3, 4\}. \quad (4.1)$$

Each x_i can only take one value, so every word already ends up in exactly one group, and the added requirement that each group have exactly four members is what turns this into a partition of W into four groups of four. One detail matters here: this setup labels the four groups, so assigning $\{w_1, w_2, w_3, w_4\}$ to group 1 and $\{w_5, \dots, w_8\}$ to group 2 counts as a different assignment from swapping those two labels, even though both describe the same grouping of words. The number of assignments satisfying C is therefore $16!/(4!)^4 = 63,063,000$, but the puzzle itself does not care which number a group is called, so the number of genuinely distinct groupings is smaller: dividing out the $4!$ ways to relabel the four groups gives $16!/((4!)^4 \cdot 4!) = 2,627,625$, the number used later in this thesis.

5 Evaluation and Discussion

5.1 Experiments

How do we build it: data, pipeline, modules.

- Pure embedding + clustering solver
- Pure LLM solver
- Embedding + LLM solver
- Improvement

6 Conclusion and Future Work

Bibliography

- [1] *Solving Wordle Using Information Theory*. Feb. 6, 2022. URL: <https://www.youtube.com/watch?v=v68zYyaEmEA> (visited on 07/30/2026).
- [2] The New York Times. *Connections (2026-01-05)*. 2026. URL: <https://www.nytimes.com/games/connections/2026-01-05> (visited on 01/09/2026).
- [3] The New York Times. *Connections (2026-01-03)*. 2026. URL: <https://www.nytimes.com/games/connections/2026-01-03> (visited on 01/09/2026).
- [4] The New York Times. *Connections (2025-12-29)*. 2025. URL: <https://www.nytimes.com/games/connections/2025-12-29> (visited on 01/09/2026).
- [5] The New York Times. *Connections Tips and Tricks*. Nov. 2023. URL: <https://www.nytimes.com/2023/11/06/crosswords/connections-tips-and-tricks.html> (visited on 01/09/2026).
- [6] Angel Yahir Lored Lopez, Tyler McDonald, and Ali Emami. “NYT-Connections: A Deceptively Simple Text Classification Task that Stumps System-1 Thinkers”. In: *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*. Abu Dhabi, UAE: Association for Computational Linguistics, 2025, pp. 1952–1963. URL: <https://aclanthology.org/2025.coling-main.134/> (visited on 12/17/2025).
- [7] Robert Geirhos et al. “Shortcut learning in deep neural networks”. In: *Nature Machine Intelligence* 2.11 (2020), pp. 665–673. DOI: 10.1038/s42256-020-00257-z. URL: <https://www.nature.com/articles/s42256-020-00257-z> (visited on 11/18/2025).
- [8] Graham Todd et al. “Missed Connections: Lateral Thinking Puzzles for Large Language Models”. In: *2024 IEEE Conference on Games (CoG)*. 2024, pp. 1–8. DOI: 10.1109/CoG60054.2024.10645557. URL: <https://ieeexplore.ieee.org/document/10645557> (visited on 01/05/2026).
- [9] Emily Zhang, Yanan Jiang, and Peixuan Ye. *Learning Semantic Complexities of NYT Connections*. CS224n course final report. Stanford University. URL: <https://web.stanford.edu/class/cs224n/final-reports/256847963.pdf>.

Bibliography

- [10] Tomas Mikolov et al. *Distributed Representations of Words and Phrases and their Compositionality*. 2013. arXiv: 1310.4546 [cs]. URL: <http://arxiv.org/abs/1310.4546> (visited on 11/23/2025).
- [11] Dhivya Chandrasekaran and Vijay Mago. “Evolution of Semantic Similarity—A Survey”. In: *ACM Computing Surveys* 54.2 (2021), 41:1–41:37. DOI: 10.1145/3440755. URL: <https://doi.org/10.1145/3440755> (visited on 11/18/2025).
- [12] Francesca Incitti, Federico Urli, and Lauro Snidaro. “Beyond word embeddings: A survey”. In: *Information Fusion* 89 (2023), pp. 418–436. DOI: 10.1016/j.inffus.2022.08.024. URL: <https://www.sciencedirect.com/science/article/pii/S1566253522001233> (visited on 11/20/2025).
- [13] Rajvardhan Patil et al. “A Survey of Text Representation and Embedding Techniques in NLP”. In: *IEEE Access* (2023). DOI: 10.1109/ACCESS.2023.3266377.
- [14] Martin Ester et al. “A density-based algorithm for discovering clusters in large spatial databases with noise”. In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD’96)*. Portland, Oregon: AAAI Press, 1996, pp. 226–231.
- [15] David Arthur and Sergei Vassilvitskii. “k-means++: the advantages of careful seeding”. In: *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA ’07)*. 2007, pp. 1027–1035. ISBN: 978-0-89871-624-5.
- [16] Chen Huang and Guoxiu He. *Text Clustering as Classification with LLMs*. 2025. arXiv: 2410.00927 [cs]. URL: <http://arxiv.org/abs/2410.00927> (visited on 11/18/2025).
- [17] Alina Petukhova, João P. Matos-Carvalho, and Nuno Fachada. “Text clustering with large language model embeddings”. In: *International Journal of Cognitive Computing in Engineering* 6 (2025), pp. 100–108. DOI: 10.1016/j.ijcce.2024.11.004. URL: <https://www.sciencedirect.com/science/article/pii/S2666307424000482> (visited on 11/18/2025).
- [18] Jason Wei et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. 2023. arXiv: 2201.11903 [cs]. URL: <http://arxiv.org/abs/2201.11903> (visited on 11/18/2025).

Bibliography

- [19] *Prompt Engineering Guide – Chain-of-Thought*. URL: <https://www.promptingguide.ai/techniques/cot> (visited on 11/23/2025).
- [20] Graham Todd et al. “Missed Connections: Lateral Thinking Puzzles for Large Language Models”. In: *2024 IEEE Conference on Games (CoG)*. 2024 IEEE Conference on Games (CoG). Aug. 2024, pp. 1–8. DOI: 10.1109/CoG60054.2024.10645557. URL: <https://ieeexplore.ieee.org/document/10645557> (visited on 01/05/2026).
- [21] Prisha Samadarshi et al. *Connecting the Dots: Evaluating Abstract Reasoning Capabilities of LLMs Using the New York Times Connections Word Game*. Oct. 14, 2024. DOI: 10.48550/arXiv.2406.11012. arXiv: 2406.11012 [cs.CL]. URL: <http://arxiv.org/abs/2406.11012> (visited on 07/02/2026). Pre-published.
- [22] Angel Yahir Loredó Lopez, Tyler McDonald, and Ali Emami. “NYT-Connections: A Deceptively Simple Text Classification Task That Stumps System-1 Thinkers”. In: *Proceedings of the 31st International Conference on Computational Linguistics*. COLING 2025. Ed. by Owen Rambow et al. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 1952–1963. URL: <https://aclanthology.org/2025.coling-main.134/> (visited on 12/17/2025).
- [23] Emily Zhang, Yanan Jiang, and Peixuan Ye. *Learning Semantic Complexities of NYT Connections*. URL: <https://web.stanford.edu/class/cs224n/final-reports/256847963.pdf>.
- [24] *Constraint Satisfaction Problem*. In: *Wikipedia*. Jan. 21, 2026. URL: https://en.wikipedia.org/w/index.php?title=Constraint_satisfaction_problem&oldid=1334133666 (visited on 07/31/2026).
- [25] Sally C. Brailsford, Chris N. Potts, and Barbara M. Smith. “Constraint Satisfaction Problems: Algorithms and Applications”. In: *European Journal of Operational Research* 119.3 (Dec. 16, 1999), pp. 557–581. ISSN: 0377-2217. DOI: 10.1016/S0377-2217(98)00364-6. URL: <https://www.sciencedirect.com/science/article/pii/S0377221798003646> (visited on 07/31/2026).
- [26] José Enrique Gallardo, Carlos Cotta, and Antonio José Fernández. “Solving Weighted Constraint Satisfaction Problems with Memetic/Exact Hybrid Algorithms”. In: *Journal of Artificial Intelligence Research* 35 (July 28, 2009), pp. 533–555. ISSN: 1076-

Bibliography

9757. DOI: 10.1613/jair.2770. arXiv: 1401.3477 [cs.AI]. URL: <http://arxiv.org/abs/1401.3477> (visited on 07/31/2026).
- [27] Matthew L. Ginsberg. *Dr.Fill: Crosswords and an Implemented Solver for Singly Weighted CSPs*. Jan. 18, 2014. DOI: 10.1613/jair.3437. arXiv: 1401.4597 [cs.AI]. URL: <http://arxiv.org/abs/1401.4597> (visited on 07/31/2026). Pre-published.
- [28] *Partition of a Set*. In: *Wikipedia*. May 5, 2026. URL: https://en.wikipedia.org/w/index.php?title=Partition_of_a_set&oldid=1352723986 (visited on 07/31/2026).
- [29] Tobias Müller. “Solving Set Partitioning Problems with Constraint Programming”. In: *Proceedings of the Sixth International Conference on the Practical Application of Prolog and the Fourth International Conference on the Practical Application of Constraint Technology (PAPPACT98)*. London, UK: The Practical Application Company Ltd., 1998, pp. 313–332. URL: https://www.ps.uni-saarland.de/Publications/documents/Mueller_98a.pdf (visited on 07/31/2026).
- [30] *Information Theory*. In: *Wikipedia*. July 26, 2026. URL: https://en.wikipedia.org/w/index.php?title=Information_theory&oldid=1366101152 (visited on 08/02/2026).
- [31] C. E. Shannon. “A Mathematical Theory of Communication”. In: *The Bell System Technical Journal* 27.3 (July 1948), pp. 379–423. ISSN: 0005-8580. DOI: 10.1002/j.1538-7305.1948.tb01338.x. URL: <https://ieeexplore.ieee.org/document/6773024> (visited on 08/02/2026).
- [32] Henry Pinkard and Laura Waller. *A Visual Introduction to Information Theory*. Mar. 6, 2026. DOI: 10.48550/arXiv.2206.07867. arXiv: 2206.07867 [cs.IT]. URL: <http://arxiv.org/abs/2206.07867> (visited on 08/02/2026). Pre-published.
- [33] Talal Aladaileh et al. “Solving Wordle Using Information Theory”. In: *Northeast Journal of Complex Systems (NEJCS)* 8.1 (Apr. 6, 2026). ISSN: 2577-8439. DOI: 10.63562/2577-8439.1146. URL: <https://orb.binghamton.edu/nejcs/vol8/iss1/6>.

Bibliography

- [34] Michael Cunanan and Michael Thielscher. *On Optimal Strategies for Wordle and General Guessing Games*. May 16, 2023. DOI: 10.48550/arXiv.2305.09111. arXiv: 2305.09111 [cs.AI]. URL: <http://arxiv.org/abs/2305.09111> (visited on 08/05/2026). Pre-published.